

Data Structures & Algorithms

Name - Piyali Sahoo

Roll no - UGT04/BTCSE/2025/016

1. (b) Data and a pointer to the next node,
2. (c) NULL
3. (b) self-referential structure,
4. (b) head $\neq$ NULL
5. (b) newnode $\rightarrow$ next = head; head = newnode;
6. (b) temp = head; head = head $\rightarrow$ next; free(temp);
7. (c) Traversing from tail to head.
8. (b) n

Name - Piyali Sahoo

Roll No - UGT04/BTCSE/2025/016

9. (c) Copying the data of p $\rightarrow$ next into p, then deleting p $\rightarrow$ next
10. (a) ~~while (p $\rightarrow$ next $\neq$ NULL)~~ (b) while (p != NULL)
11. (b) p $\rightarrow$ next = NULL
- 12.
13. (c) Three.
14. (b) Exactly one node.
15. (b) P = (struct node*) malloc(size of (struct node));
16. (b) Avoid special case handling for insertion and deletion at the head
17. (c) The first node.

Name - Piyali Sahoo

Roll no - UGT04/BTCSE/2025/016

Name - Piyali Sahoo

Roll NO - UGT/04/BTCSE/2025/016

26. struct node * reverse

(struct node * head){

struct node * prev = NULL;

struct node * curr = head;

struct node * next = NULL;

while (curr != NULL){

next = curr → next;

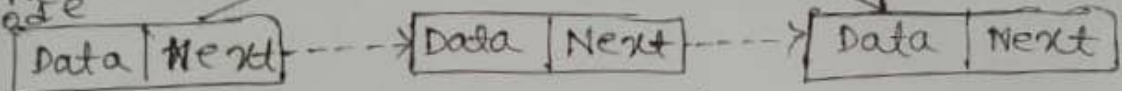
curr → next = prev;

prev = curr;

curr = next;

} return prev;

27. previous node



pointer adjustment

prev → next = del → next;

free(del)

Name - Piyali Sahoo

Roll NO - UGT/04/BTCSE/2025/016

28. 5 → 8 → 12 → 3 → NULL

7 → 5 → 8 → 12 → 3 → NULL

7 → 5 → 8 → 3 → 9 → NULL

Final → 7 → 5 → 8 → 3 → 9 → NULL

Name - Piyali Sahoo

Roll NO - UGT/04/BTCSE/2025/016

Name - Piyali Sahoo

Roll NO - U01041BTCSE/2025/016

18. (b) It does not support random (indexed) access.

19. (b) Loss of access to all nodes, causing a memory leak.

20. (b) A slow pointer, and a fast pointer.

21. (c) The last node.

22. (b) Nodes may be scattered anywhere in memory.

23.

```
#include <stdio.h>
#include <stdlib.h>
struct node{
    int data;
    struct node *next;};
```

Name - Piyali Sahoo

Roll NO - U01041BTCSE/2025/016

24. 30, 50, 70

25.

```
void insertEnd (struct node ** head, int x){
    struct node t = (struct node) malloc
```

```
    (sizeof (struct node));
    t->data = x;
```

```
    t->next = NULL;
```

```
    if (*head == NULL){
```

```
        *head = t;
```

```
        return;
```

```
    }
```

```
    struct node * p = *head
```

```
    while (p->next != NULL)
```

```
        p = p->next;
```

```
    p->next = t;
```

Name - Piyali Sahoo

Roll NO - U01041BTCSE/2025/016